Single-Label Multi-Modal Field of Research Classification

Universität Zürich[UZH]

DIS Dynamic and Distributed Information Systems

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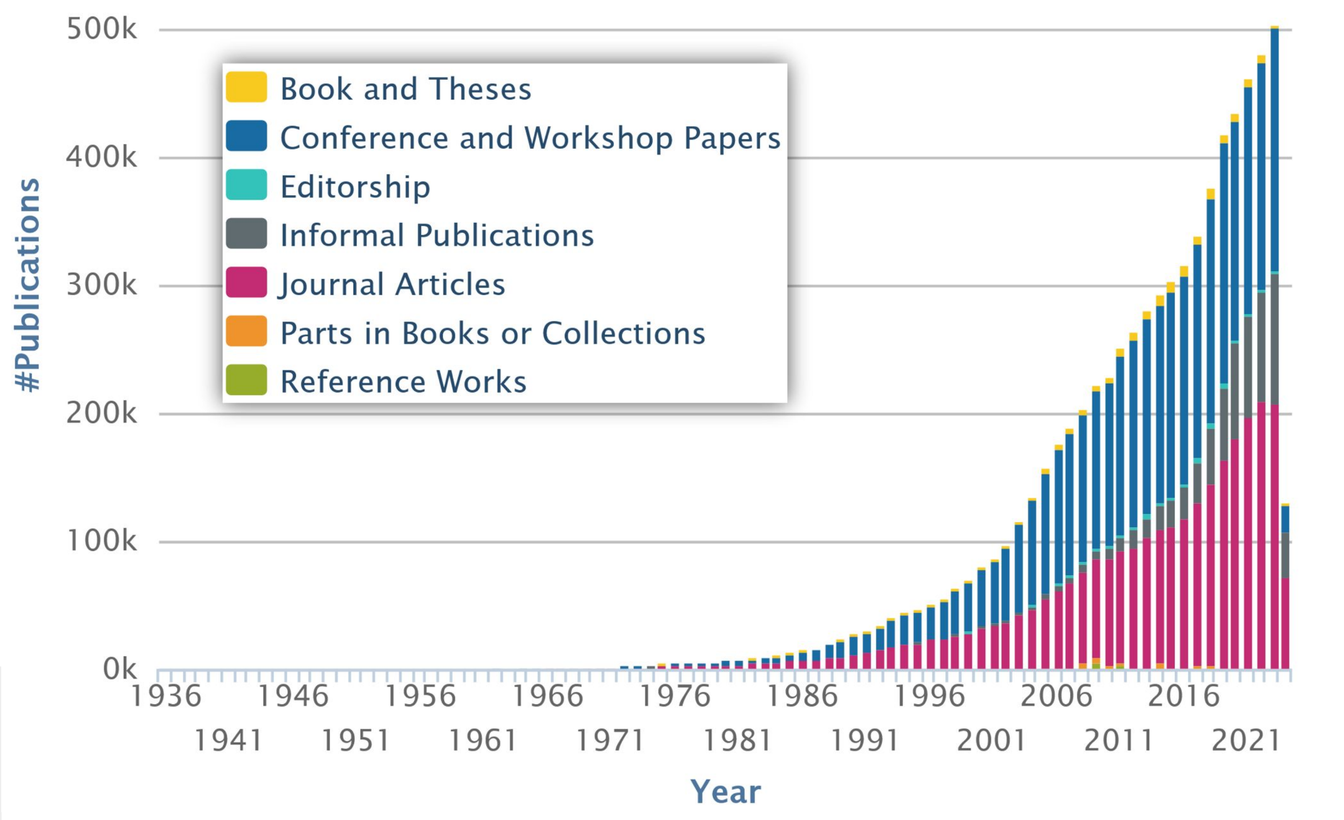
Motivation

- Exponential increase in the publication of scientific records.
- Automatic annotation systems are crucial for human annotators working at digital libraries to annotate the research papers into classes.
- Our objective is to **classify scientific articles for their respective field of research** of the 123 pre-defined hierarchical classes from the ORKG taxonomy of research fields.

Our method **ranked first** in the shared task Field of Research Classification of Scholarly Publications of the 1st **Workshop on Natural Scientific Language Processing and Research Knowledge Graphs** (NSLP 2024).



Source: https://dblp.org/statistics/publicationsperyear.html

SLAMFORC System

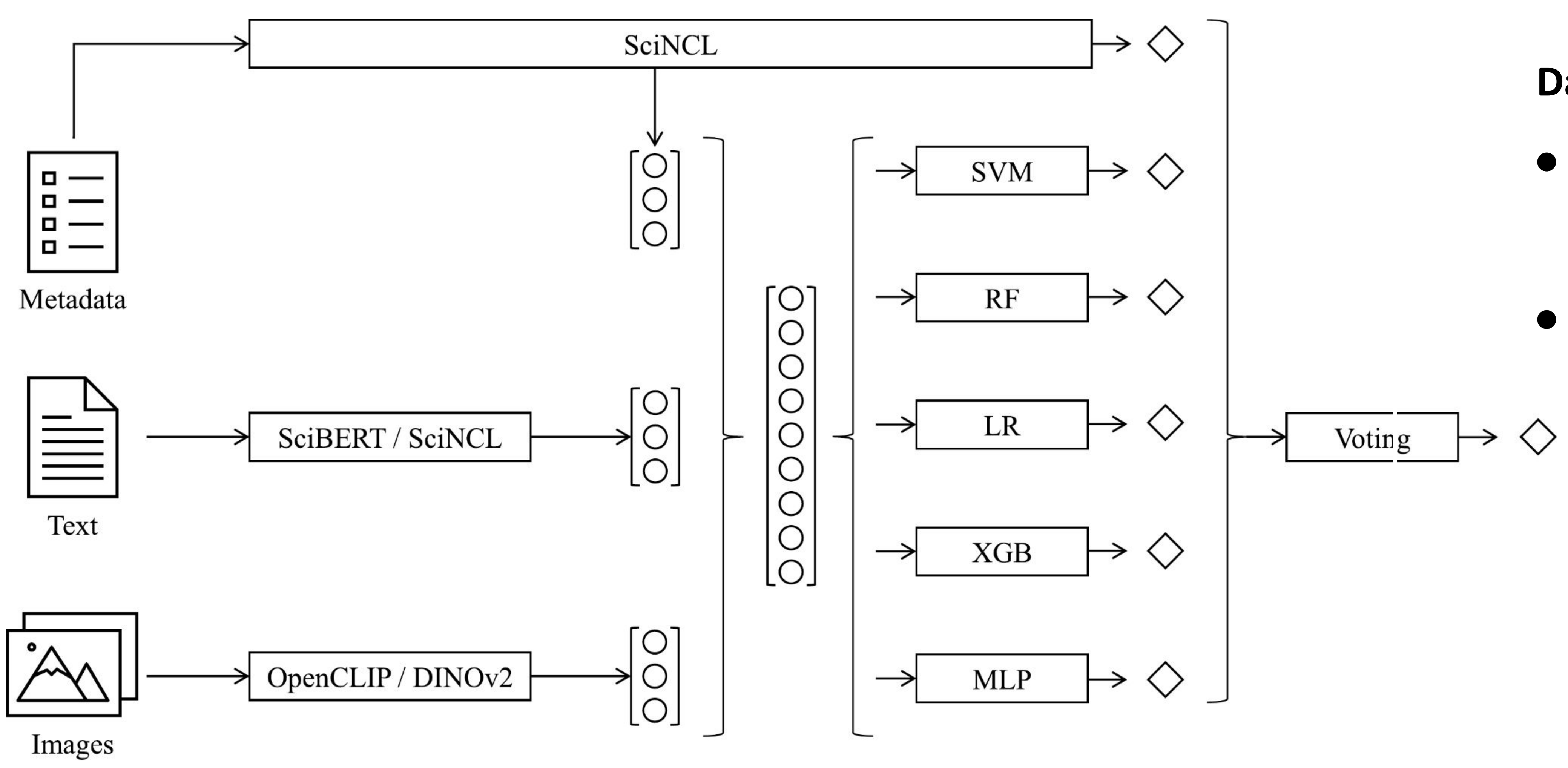


Figure: Overview of the SLAMFORC system showing the integration of multi-modal data and classifiers.

SLAMFORC System: mixture of traditional classifiers including SVM, Random Forest, Logistic Regression and eXtreme Gradient Boosting and neural methods including MLP, SciNCL that we combined with an ensemble voting method.

***Note**: SciNCL is a pre-trained BERT language model to generate document-level embeddings of research papers*

Dataset : 59,500 English scientific papers

- Metadata fields: title, authors, abstract, DOI, URL to the full paper (if available), publisher, publication month, and year
- Classes are 123 fields of research (FoR) across five major domains and four hierarchical levels, with mapping to the ORKG taxonomy

SLAMFORC System: Multi-Modal Data

- **Crossref**: enhance the text data of papers (e.g., retrieved the annotated subjects)
- **Metadata**: used title, abstract, and publisher information from given dataset
- **Full text**: retrieved full text of document if available using PaperMage [19]
- **Visual Information**: OpenCLIP [11] model pre-trained on the LAION-5B dataset [26], a pre-trained DINOv2 [22] model

Results

- Our system achieved the **highest precision and F1** and the **second-best accuracy and recall** values of all submissions.
- Our feature ablation study indicates that the meta information is more contributing towards the classification.
- Our combined approach is better than the end-to-end LLM (SciNCL) solution on only the metadata.
- We trained several SOTA classifiers that could handle vectors as input and predict the single-label class as outcome.

Table: Ablation Study Results.

Classifier	Accuracy	Precision	Recall	F1
Support Vector Machine	0.755	0.754	0.755	0.752
Random Forest	0.755	0.754	0.755	0.753
Logistic Regression	0.750	0.755	0.750	0.750
XGBoost	0.731	0.735	0.731	0.731
Multilayer Perceptron	0.743	0.748	0.743	0.743

*Table: The final evaluation results on the test set as measured by accuracy, weighted precision, recall, and F1 (best for each in **bold**, runner-up underlined). Our submission is the first line. The F1 scores are sorted in descending order, indicating that higher scores are listed first.*

ID	Accuracy	Precision	Recall	F1 ↓
SLAMFORC	0.756	**0.757**	0.756	**0.754**
688995	0.754	0.755	0.754	0.752
687946	**0.757**	0.754	**0.757**	0.750
689747	0.748	0.744	0.748	0.743
688510	0.743	0.742	0.743	0.739
651741	0.730	0.725	0.730	0.724
649383	0.726	0.719	0.726	0.720
686435	0.706	0.703	0.706	0.693
686384	0.702	0.695	0.702	0.692
689251	0.682	0.679	0.682	0.678
689454	0.658	0.659	0.658	0.653
647796	0.058	0.061	0.058	0.057
678150	0.004	0.002	0.004	0.002

Concluding Remarks:

- Evaluation Method: need an evaluation metric that integrates semantic analysis within the taxonomy to better understand system performance and address misclassifications.

More Info: FoRC: Field of Research Classification of Scholarly Publications (Subtask I)
Workshop: https://github.com/NFDI4DS/nslp2024/tree/main